



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Certified Lean Six Sigma Yellow Belt (CLSSYB) Training

Course Code: C524

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 5

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1	1 June 2026	Initial release — CLSSYB 2-day lesson plan.	Dr. Alfred Ang
2	15 June 2026	Refreshed labs against the CSSC Yellow Belt reference.	Dr. Alfred Ang
3	1 July 2026	Aligned the schedule with the Contoso Service Desk case-study scenario.	Dr. Alfred Ang
4	19 July 2026	Major content expansion: course restructured to follow the DMAIC roadmap across six topics with 14 labs.	Dr. Alfred Ang
5	19 July 2026	Converted to the non-funded commercial course standard; the course code is now C524. The 165 minutes previously reserved for formal evaluation on Day 2 are returned to teaching — elective labs 11-14 are now delivered in class. Rebuilt from a single source (course_data.py + data_domain1-6.py) via the non-wsj-courseware-build engine.	Dr. Alfred Ang

TABLE OF CONTENTS

Course Information.....	4
Learning Outcomes.....	4
Learning Reinforcement.....	4
Course Schedule.....	4
Day 1 — Six Sigma foundations, Define phase and process mapping.....	4
Day 2 — Measure, Analyse, Improve and Control.....	5
Lab Reference (aligned to course topics).....	6

Course Information

Course Title	Certified Lean Six Sigma Yellow Belt (CLSSYB) Training
Course Reference	C524
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	2 days · 8 training hours per day (16 hours)
Daily Timing	9:30 am – 6:30 pm (1-hour lunch; tea breaks within training time)
Mode	Instructor-led, hands-on Lean Six Sigma labs using the Contoso Service Desk scenario
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Define a project and establish the scope of work using Six Sigma's Define, Measure and Analyse phases.
- LO2: Apply Lean and Six Sigma concepts — value, waste, defects and variation — to a work process.
- LO3: Map a process using SIPOC, process maps and value stream maps to expose handoffs and waste.
- LO4: Collect and analyse process data using check sheets, Pareto charts, run charts and basic metrics.
- LO5: Identify root causes using 5 Whys, Fishbone analysis and evidence-based prioritisation.
- LO6: Recommend improvement and control actions to sustain gains, and prepare for certification.

Learning Reinforcement

- Every topic is followed by hands-on labs that learners complete on their own machine.
- Each lab ends with a 'Test it' verification step so learners can confirm their own progress.
- The trainer circulates during lab time to give individual feedback and unblock learners.
- Each topic closes with a recap mapped back to the course learning outcomes.

Course Schedule

Day 1 — Six Sigma foundations, Define phase and process mapping

Time	Duration	Topic / Activity
9:30–10:00	30 min	Welcome, course introduction and ground rules
10:00–11:15	75 min	FOUNDATIONS — What is Quality; What is Lean; What is Six Sigma; Lean Six

		Sigma; belt roles; the DMAIC roadmap
11:15–11:30	15 min	Tea break
11:30–12:30	60 min	Hands-on: Lab 1: Yellow Belt Role, Certification Paths, and Improvement Scenario
12:30–13:30	60 min	Lunch break
13:30–15:00	90 min	DMAIC · DEFINE — Voice of the Customer; VOC to CTQ translation; project charter; problem statement; scope; SIPOC; process mapping
15:00–15:15	15 min	Tea break
15:15–18:00	165 min	Hands-on: Lab 2: Lean, Six Sigma, Waste, Voice of Customer, and Value; Lab 3: SIPOC, Process Mapping, Handoffs, and SME Support; Lab 4: PDCA Small Improvement Project Charter; Lab 5: DMAIC Overview, Problem Statement, Scope, and Stakeholders; Lab 11: Affinity Diagram and Kano Analysis
18:00–18:30	30 min	Day 1 recap and Q&A

Total training time: 480 minutes (8 hours).

Day 2 — Measure, Analyse, Improve and Control

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 1 recap
9:45–11:00	75 min	DMAIC · MEASURE — the eight wastes (DOWNTIME); data types; data collection plan; check sheets; yield, DPU, DPO and DPMO; sigma level
11:00–11:15	15 min	Tea break
11:15–12:30	75 min	Hands-on: Lab 6: Data Collection, KPIs, Check Sheets, and Basic Metrics; Lab 12: Value Stream Map and Takt Time
12:30–13:30	60 min	Lunch break
13:30–14:15	45 min	DMAIC · ANALYZE — common vs special cause variation; Pareto; run charts; 5 Whys; Fishbone; evidence testing
14:15–15:15	60 min	Hands-on: Lab 7: Pareto, Run Charts, Variation, Yield, DPU, and DPMO; Lab 8: Root Cause Analysis with 5 Whys, Fishbone, and Evidence
15:15–15:30	15 min	Tea break
15:30–16:00	30 min	DMAIC · IMPROVE — solution generation and

		selection; 5S; poka-yoke; standard work; kaizen; piloting
16:00–17:00	60 min	Hands-on: Lab 9: Countermeasures, 5S, Mistake Proofing, Standard Work, and Kaizen; Lab 13: Solution Selection Matrix, Benchmarking, and FMEA
17:00–17:30	30 min	DMAIC · CONTROL — control plans; process control; visual management; SOPs; huddles; A3 and handover
17:30–18:15	45 min	Hands-on: Lab 10: Control Plan, A3 Summary, Handover, and Certification Readiness; Lab 14: Descriptive Statistics and Implementation Planning
18:15–18:30	15 min	Course recap, consolidation and next steps

Total training time: 480 minutes (8 hours).

Lab Reference (aligned to course topics)

Topic	Coverage	Labs
Topic 01: Six Sigma Foundations	15%	Lab 1
Topic 02: Define — Scope the Problem	25%	Lab 2, Lab 3, Lab 4, Lab 5, Lab 11
Topic 03: Measure — Quantify Performance	25%	Lab 6, Lab 12
Topic 04: Analyze — Find the Root Cause	20%	Lab 7, Lab 8
Topic 05: Improve — Fix the Cause	10%	Lab 9, Lab 13
Topic 06: Control — Hold the Gain	5%	Lab 10, Lab 14